Einstein Gaps on Four-Manifolds

Constructions, compatibility estimates, and current sharpness problems

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1 Mathematical setting

The common problem is to determine how closely a smooth four-manifold can approach Einstein geometry, under specified topological and actual gauge-field constraints. The work concerns uniform lower bounds, optimal values, geometric constructions, and the distinction between a vanishing infimum and an exact Einstein metric. These questions are related but not equivalent.

This synthesis records the mathematical contributions of Sessions S1003A01 and S1003A02. It separates their inherited theory from new constructions and estimates, and separates conclusions about all competing metrics

from conclusions about marked cells or restricted geometric classes. The full proof texts and computational sources accompany it. The snapshot was taken on 7 October 2026 at 10:47:26–10:47:27, Asia/Shanghai; the latest included research notes are A01 Step 38 and A02 Step 44.

On a closed oriented four-manifold set

$$P(g) = \int_M |\operatorname{Ric}^\circ|^2, \quad S(g) = \int_M s^2, \quad R(g) = P(g) + \tfrac{1}{4}S(g), \quad T_\pm = 2\chi \pm 3\tau.$$

With the operator norm convention for Weyl curvature,

$$4\pi^2 T_\pm = \frac{S}{24} + 2\int |W^\pm|^2 - \frac{P}{2}.$$

For $R > 0$, the normalized defects are

$$\mathcal{J}(g) = \frac{4P}{R}, \qquad \mathcal{E}(g) = 4 - \frac{(\int s)^2}{VR} = \mathcal{J}(g) + \frac{S - (\int s)^2/V}{R}.$$

Write $\alpha(M) = \inf_g \mathcal{E}(g)$. A Ricci-flat metric has $\mathcal{E} = 4$, while $\mathcal{J}$ is undefined there. On a constant scalar-curvature metric with $R > 0$, the two defects agree.

For surgery families,

$$M = X\#k\overline{\mathbb{CP}}^2\#\ell(S^1 \times S^3), \qquad A = c_1^2(X) > 0, \quad Q = k + 4\ell, \quad h = Q/A.$$

The original balanced problem includes both blow-ups and handles. A theorem for pure blow-ups $\ell = 0$ is not automatically a theorem for that problem.

Local raw energies use a different notation:

$$J_\lambda(g) = \int_C |\operatorname{Ric} - \lambda g|^2.$$

Thus the notes' J_6, on a background $\operatorname{Ric} = -6g$, means $\int |\operatorname{Ric} + 6g|^2$, not the normalized $\mathcal{J}$. Numerical cell costs are usually divided by $8\pi^2$.

2 Inherited analytical backbone

The following are inputs to these Sessions, not newly established results of the two runs:

- The single-solution finite-parameter estimates and the N1 normalized gap, including the explicit negative-class-square and class-square-shortfall versions.
- The I40/I41 coercivity estimate

$$\Sigma \le C\{\Pi + \sqrt{\Pi Y}\}, \quad \Sigma = S - 32\pi^2 p, \quad \Pi = P + D_2, \quad Y = (T_+ - a^2)_+.$$

- The matched linear scale $ch \le \alpha \le 4h/(1+h)$ on the specified balanced families, and the small-h variational reduction

$$\alpha_j = 4\frac{\inf_g P(g)}{8\pi^2 A_j} + O(h_j^2).$$

- The finite-topological-density endpoint theorem, with a constant depending on an upper bound for χ/A.
- The all-metric pure-handle sharp value, classical monopole suppliers, and the previous relative Chern–Weil and complete-cell transfer framework.

The full inherited manuscript is Curvature Realizability. Its editable sections and source library are also included. Material copied back into the Skill on 4 October, notably the strict cheap-cell construction, remains an output of these Sessions rather than an older input.

The relevant classical distinction is between $\inf R$ and $\inf P$. LeBrun's exact Ricci-energy calculations and recovery constructions do not allow subtraction of independently attained infima. Likewise, the Malchiodi–Malizia Weyl-energy improvement does not by itself control the additional scalar contribution to P or to a raw Einstein residual.

3 A strict improvement of the balanced asymptotic upper bound

3.1 The complete cheap cell

A01 and A02 construct and check a smooth ordinary (-1)-blow-up cell with an exact complex-hyperbolic outer collar. The useful geometry is triaxial and perpendicular to the usual Kähler alignment. The original argument checks positive definiteness, bolt regularity, ordinary blow-up topology relative to the marked boundary, and finite-order contact with the CH metric. A further cutoff smoothing has arbitrarily small energy loss.

The current degree-20 profile, with $x_0 = 3/100$ and outer parameter $X = 128$, has the recorded interval enclosure

$$\frac{J_6}{8\pi^2} \in [0.995670030344086466\ldots, 0.995670030355130639\ldots].$$

In particular there is a fixed $\eta > 0$ such that the rescaled smooth cell has cost at most $8\pi^2(1-\eta)$. The analytic admissibility and smoothing argument is separate from the interval integration of a specific profile.

For appropriately selected residual-finite covers, sufficiently many disjoint marked balls admit insertion of this cell. The handle pieces are chosen at shrinking scales so that their aggregate errors are $o(Q)$. For the balanced ratios $k/Q \to 2/9$, $\ell/Q \to 7/36$,

$$\limsup \frac{\alpha(M_j)}{h_j} \le 4 - \frac{8}{9}\eta < 3.9961512 < 4.$$

The bookkeeping is $28/9 + (8/9)p_{1,0}$, not $4p_{1,0}$, where $p_{1,0}$ is the normalized single-blow-up cost.

This changes the answer to a substantive question: the standard shrinking recovery is not asymptotically optimal on these selected balanced families. It does not identify the sharp coefficient or establish the same bound for every prescribed covering tower. Refining the interval enclosure is a strengthening of the same construction, not a second independent mechanism.

Proofs: A01 Step 10, A01 Step 11, A02 independent cell calculation, and A01 Step 33, section 6.

3.2 Transfer to non-ball-quotient Kähler–Einstein backgrounds

A01 subsequently transfers the construction to compactifications of ramified arithmetic ball-quotient covers. Their smooth negative KE metrics have

$$A_m = c_1^2(\bar{X}_m), \qquad \Delta_m = 3\chi(\bar{X}_m) - A_m > 0, \qquad \Delta_m/A_m \longrightarrow 0.$$

The strict Chern-number inequality shows that these compact surfaces are not compact ball quotients.

The argument uses smooth convergence on a fixed compact region away from the compactification divisor. Galois lifts provide $d_m \asymp A_m$ disjoint insertion regions with the same C^2 error $\varepsilon_m \to 0$. A fixed annular buffer joins the exact CH cell to the KE host, at raw Ricci-defect cost $O(\varepsilon_m^2)$ per insertion. Across $O(Q_m)$ insertions this is $o(Q_m)$; the unchanged KE bulk contributes no raw Einstein defect.

If $\Delta_m/Q_m \to r \in [0,1)$ and the surgery parameters are balanced,

$$\limsup \frac{\alpha(M_m)}{h_m} \le 4 - \frac{8\eta}{9}(1-r),$$

while the inherited actual-field theorem supplies a positive linear lower bound. This is a new family and transfer argument, not a second cheap-cell construction.

More generally, put

$$\theta = k/Q, \qquad r = \Delta/Q, \qquad \sigma = 4r + 18\theta.$$

The characteristic numbers give

$$T_+ = A - Q, \qquad \frac{3T_-}{A} = 1 + (\sigma - 3)h.$$

On the supplied families the existing lower theory and the new recovery give

$$\liminf \frac{\alpha}{h} \geq \max\left\{\frac{4}{3}(3-\sigma)_+, \frac{18}{19}(4+c_0-\sigma)_+\right\}, \qquad \limsup \frac{\alpha}{h} \leq 4(1-\eta\theta).$$

The lower information depends on σ; this separated recovery depends on θ. Neither formula identifies the sharp transition or minimum. The loss of the balanced saving at $r = 1$ is not a general existence barrier.

Proofs: ramified transfer and slope law. These results use a fixed region away from the divisor; they do not depend on the later, incomplete near-divisor Calabi-cap transfer.

4 Endpoint estimates without a density bound in a specified region

For an actual class, let

$$D_\beta = (1+\beta^{-1})S + 6(\beta+1)\int |W^+|^2 - 72\pi^2 p, \quad \kappa_\beta = \fint |\operatorname{Ric}+g|^2 + D_\beta/V.$$

Fix $\vartheta < 3/4$. A02's L-prime statement assumes a fixed Spin^c structure supplying an unperturbed monopole on every metric of the chosen conformal class, with

$$p \geq A > 0, \qquad a^2 \geq T_+, \qquad T_+ \leq \vartheta A.$$

It proves

$$\kappa_\beta(g) \geq c_L(\vartheta) > 0 \quad \text{for every metric in that class and every finite } \beta \geq 2.$$

The same constant works on the entire finite half-line; this is not a bound only on a compact parameter interval. There is no Euler-density assumption.

The large-parameter part follows by combining Chern–Weil with LeBrun's weighted estimate. The decisive numerator is

$$N_\vartheta(\beta) = (3/4-\vartheta)\beta^2 + (1/2-11\vartheta/4)\beta + 6 - \vartheta,$$

with discriminant

$$\frac{57}{16}(\vartheta - 2/3)(\vartheta + 142/19).$$

At $\vartheta = 2/3$, it equals $(\beta-8)^2/12$. For $2/3 \leq \vartheta < 3/4$, the inherited finite-parameter non-saturation theorem handles the bounded bad interval, and a conformal comparison returns to the original metric. The constant degenerates as $\vartheta \uparrow 3/4$.

For a minimal general-type Kähler surface with $p_g \geq 1$, sign averaging and a split according to the blow-up ratio combine this result with the finite-density theorem. The resulting Corollary K gives, on all its blow-ups,

$$\kappa_\beta(g) \geq c_K(\beta-2)^2/\beta^2$$

for all metrics and all $\beta \geq 2$. This expands the actual endpoint range. The unrestricted high-density endpoint remains open, and a bound for κ_β is not a bound for α: field alignment and Einstein proximity measure different conditions.

Proof entry: A02 Step 6, with the full conformal argument in Step 5 verifier integration.

5 General background response and conformal selection

5.1 Burns insertion on an arbitrary fixed background

A02's Theorem F takes a closed oriented smooth Riemannian four-manifold (M, g), a point p, and a unit self-dual form $\zeta \in \Lambda^+_p$. For a Burns insertion with exceptional area πx and cutoff radius $x^{1/5}$, it gives

$$P(\widehat{g}_x) = P(g) + 8\pi^2\{1 - xR^+_p(\zeta,\zeta)\} + O(x^{6/5}), \qquad R^+ = W^+ + \frac{s}{12}I.$$

The remainder constant depends on local background control. The corresponding raw Einstein-residual formula requires an Einstein background.

The proof combines the scalar-flat, self-dual-flat linear Burns tail with its cut-annulus transgression and separate estimates on the core, intermediate region, and cutoff region. It identifies the first-order dependence on ambient curvature and insertion direction. For a negative KE background, the perpendicular directions lie in $\ker R^+$; on a real hyperbolic background every direction has positive first-order cost.

This is an upper construction, not a lower bound for the infimum over all cells. Its remainder is larger than a possible $x^2 \log(1/x)$ contribution, so it does not prove the more delicate kernel-direction saving uniformly on general backgrounds.

Proof: A02 Step 20.

5.2 Conformal selection without a symmetry assumption

A01 treats every smooth complete connected cell that is exactly CH outside a compact set, with normalization $\mathrm{Ric} = -6g$. No sign condition is imposed on the scalar curvature in the core. Ordered global barriers, exhaustion, and the maximum principle give a unique positive solution

$$-6\Delta_g u + s_g u + 24u^3 = 0, \qquad u \to 1,$$

so that $\widehat{g} = u^2 g$ has scalar curvature -24. For every $4 < m < 2 + 2\sqrt{3}$, its derivatives decay as $O(e^{-mr})$. This removes the linear boundary transgression and yields

$$P(g) - P(\widehat{g}) = 24 \int |\nabla u|^2 + 3 \int (\Delta u/u)^2.$$

Together with cutoff recovery, it proves equality of the exact-CH-ended P- and J_6-infima and the corresponding scalar-normalized aCH infimum for $2 < \mu < 2 + 2\sqrt{3}$.

This removes the diagonal, symmetry, and scalar-sign assumptions from that local variational reduction. It does not prove attainment or determine the unrestricted cell value. Arbitrary asymptotically CH inputs require separate forcing and decay hypotheses.

Proof: A01 Step 22.

6 An exact Einstein filling with nonstandard conformal infinity

A02's Theorem E constructs a complete real-analytic diagonal Bianchi–IX metric on $\mathcal{O}(-1) \to \mathbb{CP}^1$ with

$$\mathrm{Ric} = -6g, \qquad \rho_\infty \in [2.19013, 2.19242].$$

The conformal infinity is a nonstandard Rossi CR sphere, and the convergence rate is at least $O(e^{-1.99t})$.

The argument consists of an analytic Fuchsian bolt construction, exact constraint propagation, interval transport of forty initial boxes, opposite exit faces of a verified block, a Wazewski continuation argument, and quadratic drift control at infinity. An off-bolt interval calculation shows that both chiral Weyl blocks have simple spectrum. Analyticity then excludes a conformally Kähler structure on any nonempty open set, in either orientation.

This is a precise existence result with a computer-assisted proof component. It is independent of the cheap-cell upper construction. Because $\rho_\infty \neq 1$, it is not a zero-cost filling with the prescribed standard CH boundary. In particular it does not show that the marked-cell infimum vanishes.

Proof and data: A02 Step 15, Rossi enclosure, Weyl discriminants.

7 The actual-field mechanism behind a saving

The newer A02 analysis works with simultaneous actual monopole solutions for the full exceptional-sign family on the same scalar-−4 metric. The input includes the required simple-type identity and, for handle summands, the appropriate higher-type or holonomy supplier. Set

$$\rho_\epsilon = |\Phi_\epsilon|^2 \in [0,2], \quad n_\epsilon = \psi_\epsilon/\rho_\epsilon, \quad \mathfrak{S} = 8\pi^2 Q - P.$$

On the nonzero-spinor set, decompose the normalized spinor derivative into amplitude, phase, Lee, and Nijenhuis components. The pointwise Dirac equation fixes the Lee component; it does not fix the Nijenhuis component. Retaining the amplitude Weitzenböck identity and the same-family Walsh terms gives

$$\mathfrak{S} = \frac{1}{2}\mathbb{E}_\epsilon \int \rho_\epsilon^2 |\nabla n_\epsilon|^2_{\mathcal{W}_2} - \mathcal{B}, \qquad \mathcal{B} \geq 0.$$

Here $\mathcal{B}$ is the explicit sum of amplitude, phase/Lee, Weyl-model, and Walsh losses in the source theorem. The equivalent Weyl expression identifies the signed interaction with $W^+ - W_{0,\epsilon}$.

Consequently, if an admissible simultaneous family is integrable on its nonzero sets, then $P \geq 8\pi^2 Q$. Conversely, a fixed proportional saving forces a comparable amount of weighted Nijenhuis energy. This relates the construction-side saving to genuine gauge fields; it does not yet bound that saving on all competing geometries.

The paired-field analysis also identifies why exceptional sign changes do not automatically add a new positive cost. The relative twisting line has class $\pm E_i$, not $\pm 2E_i$; its fixed negative square is carried by the anti-self-dual part, while amplitude and directional differences see different components. Recounting a Walsh sector as an independent positive charge would count the same information twice.

Proofs: paired-field scope and Nijenhuis and Weyl identities.

8 Boundary mechanisms and limitations of local estimates

The Sessions develop exact chiral transgressions on diagonal annuli, boundary-corrected almost-Kähler formulas, CR pinning, standard-CR truncation, exterior ground states, and joint quadratic estimates. These have three distinct uses:

1. They compute costs for specified marked boundaries and identify zero modes that carry interface data without raw curvature cost.
2. They exclude particular controlled escape channels, including specified nearly Euclidean necks and developed-Rossi-to-product connectors.
3. They identify the hypotheses needed before a local cost can be used in a global lower bound.

For example, the corrected AK expression is

$$\widehat{P}[t_0, t_1] = \int_{t_0}^{t_1} L_{AK} + B(t_1) - B(t_0).$$

A positive bulk AK term can be cancelled by the boundary term; it is not an additional independent charge. The diagonal control calibration similarly has the exact form

$$\widehat{P} = \Delta \mathfrak{b}_+ + \int L(c - c^*), \qquad L > 0.$$

Its calibrated trajectories are ASD scalar-flat trajectories. The exact calibration does not turn sampled trajectory destinations or numerically optimized path costs into uniform lower bounds for all paths.

The pure-handle marked-cell infimum is $32\pi^2\ell$, approached but not attained for $\ell > 0$. This is a useful localization of the inherited global handle theorem. Mixed-cell additivity and identification of cell specific energy with the unrestricted closed-manifold optimum remain open.

Sources: A01 AK boundary correction, A02 transgressions, A02 truncation, A02 calibration.

9 What the latest analyses establish

9.1 Pointwise combinations and exceptional periods

The constant affine combinations considered in A02 Step 40, followed by discarding known-sign terms and applying the indicated sharp Weyl Cauchy–Schwarz estimate, recover exactly the classical threshold $P/(8\pi^2) \geq k - A/3$ for pure blow-ups. Their envelope binds at $\rho = 3/2$. This excludes that specified algebraic strategy as a source of a stronger estimate; it does not exclude nonlinear pointwise inequalities, other multipliers, joint constraints among all fields, or differential methods.

Step 42 specializes to pure blow-ups with $b^+ \geq 2$ and basic classes exactly $\pm K_X$. A classical Taubes argument excludes a nowhere-zero closed self-dual common field with zero exceptional periods. It rules out an exact formal configuration, not a sequence with small periods or thin zero sets. The almost-Kähler torsion profile retains its period-angle factor; the inequality $\eta \geq 1$ is not automatic for $b^+ \geq 2$.

9.2 Cotton pairing

Step 43 computes the model divergence on the stated almost-Kähler first jets and uses the closed constant-scalar Cotton pairing. The desired second-jet bound, treatment of zeros and cutoffs, and noncircular control of the Weyl error remain missing. A conditional estimate $P \geq R_N^2/(\Lambda A)$, together with a separate saving bound, would yield a quadratic-scale lower bound. It is not an established new Einstein gap, nor is it already a statement for the original balanced family.

9.3 The newly included perturbative note

A02 Step 44 proposes a cover-uniform perturbative theorem near CH hosts and an almost-Kähler pocket estimate. They require different conclusions.

Write

$$C_+ = \frac{1}{4\pi^2}\left(\frac{S}{24} + 2\int |W^+|^2\right) = T_+ + \frac{P}{8\pi^2}.$$

The pocket calculation, with $\ell_{AK} = 8\pi K_\omega \cdot [\omega] > 0$, $\theta = \int |\nabla\omega|^2/\ell_{AK}$, and $\varphi = \mathrm{Vol}\{s^* \geq 0\}/V < 1$, gives

$$C_+ \geq \frac{\ell_{AK}^2}{128\pi^2 V}\left[3\theta^2 - 2\theta + 1 + \frac{\varphi}{1-\varphi}(1-\theta)_+^2\right] \geq \frac{\ell_{AK}^2}{128\pi^2 V}\frac{2}{3-2\varphi}.$$

It follows from a two-region convex quadratic minimization. Relating this almost-Kähler structure to the actual canonical monopole branch is a separate question.

The claimed two-sided perturbative comparison is not established by the given proof. Indeed, at $Q = 0$ the exact Nijenhuis identity is

$$-P = \tfrac{1}{2}R_N - \mathcal{B}, \qquad \mathcal{B} \geq 0.$$

This implies $-P \leq R_N/2$, not $2P \leq R_N$. The latter is used in Step 44. Its linearized SW estimate also retains the whole derivative norm while using the same derivative energy to supply a positive mass term on a zero-potential sector. A weaker estimate might be repaired by allocating part of the derivative term, but the displayed inequality does not follow as written. The spinor normalization and the uniform Hölder inverse require consistent treatment as well.

The accompanying review also gives a geometric reason to reject a general reverse comparison: a nearby non-Einstein Kähler metric can have an integrable canonical monopole direction, hence zero Nijenhuis energy but positive P. Integrability also persists under conformal scalar normalization. The possible one-sided estimate $R_N \leq CP$ is a different statement; its uniform analytic estimates still need completion.

Accordingly, the proposed theorem must not be used as a completed two-sided cover-uniform comparison. The original note and its accompanying review are both preserved. The separate pocket inequality does not depend on the incorrect reverse comparison.

Sources: Step 42, Step 43, Step 44, and its accompanying review.

10 Classical applications and computational exploration

Canonical-model resolutions and rational blow-down constructions supply zero-α examples through existing geometry. Their role here is to separate a zero infimum from attainment and from exact Einstein existence. Real-hyperbolic blow-up nonexistence uses classical entropy and rigidity theorems. A proposed sharp formula on real-hyperbolic blow-ups remains conditional on an unproved Yamabe-invariant equality. These are not new unconditional gap theorems.

Gibbons–Hawking coupled tangent models rule out an overly strong local Liouville shortcut, but are not closed-manifold O10 counterexamples. Small-bolt asymptotics, collapsed-control optimizations, higher angular variations, and Calabi-cap transports provide candidates and tests. Their floating values do not prove global lower bounds or stability.

In particular A01's latest Calabi-cap work has no certified improvement on the CH cheap cell. A coarse Step-38 response lowered the training objective but raised the independently sampled value. Neither this failure nor a finite-dimensional absence of descent proves that all cap constructions fail. The near-divisor cap transfer and packing also still need the stated uniform chart and lattice inputs.

Sources: A02 theorem package, A01 cap transfer, A01 latest response.

11 The remaining global mathematical questions

The inherited N1 and I40/I41 theorems do not wait for these questions. The two strongest unresolved upgrades have precise endpoints.

Unrestricted O10. Determine whether the original actual-field inputs exclude an endpoint sequence with unbounded topological density, or construct a closed-manifold sequence satisfying all those inputs. The new $T_+/A < 3/4$ result removes a genuine subregion from the unknown set; it does not settle the remaining high-density range.

O11. Determine the actual infimum or sharp asymptotic coefficient on the specified families. On the selected sub-four families the answer cannot be the former candidate 4. A lower bound must allow the saving already realized by the construction. A universal decomposition into independently priced standard surgery pieces would therefore be the wrong conclusion.

There are two legitimate ways to close the missing implication: a direct global estimate using actual fields and curvature equations, or a geometric description of arbitrary near-minimizers that supplies the hypotheses of a sharp local estimate, with total remainder $o(Q)$. Neither method is imposed as a prerequisite for the other. Extra pure-blow-up, period-angle, diagonal, or collar assumptions must either be supplied from the target problem or remain explicit restrictions.

The most coherent mathematical connection emerging from the two Sessions is that topology-changing cost is not a fixed additive tariff. Ambient chiral curvature, the insertion direction, nonintegrability of actual fields, and the marked boundary can change which costs are realizable. The constructions show that some savings really occur; the missing sharp lower theory must explain how far those mechanisms can go.

12 Primary literature and attribution

The bibliography is retained in the inherited paper and Session notes. The following comparisons are especially relevant to the claimed increments.

- C. LeBrun, *Ricci Curvature, Minimal Volumes, and Seiberg–Witten Theory*, arXiv:math/0003068: curvature budgets, exact Ricci-energy infima, recovery, and the exceptional-period argument.
- C. LeBrun, *Curvature Functionals, Optimal Metrics, and the Differential Topology of 4-Manifolds*, arXiv:math/0404251: the distinction between Ricci and full-curvature optimization.

- A. Malchiodi and F. Malizia, *Weyl energy and connected sums of four-manifolds*, arXiv:2505.17752: strict Weyl-energy improvement, which must be distinguished from the joint scalar and Ricci cost controlled here.
- T. Ozuch, *Higher order obstructions to the desingularization of Einstein metrics*, arXiv:2012.13316: exact Einstein desingularization obstructions, not an arbitrary-cell energy formula.
- O. Biquard and M. Herzlich, *A Burns–Epstein invariant for ACHE 4-manifolds*, arXiv:math/0111218: boundary CR geometry and characteristic identities.
- D. Calderbank and M. Singer, *Einstein metrics and complex singularities*, arXiv:math/0206229: related complete Einstein constructions with their own topology and curvature hypotheses.

The local applicability of these sources is stated at each use in the original proofs. Establishing priority for the new formulas or filling requires comparison of their exact statements, not just matching subjects.